Exercise 2: COE2263 Digital & Analog Circuits

Problem 1

Section 4–5 Logic Simplification Using Boolean Algebra

19. Using Boolean algebra techniques, simplify the following expressions as much as possible:

(a) $A(A + B)$ **(b)** $A(\overline{A} + AB)$ **(c)** $BC + \overline{B}C$
(d) $A(A + \overline{A}B)$ **(e)** $A\overline{B}C + \overline{A}BC + \overline{A}\,\overline{B}C$

20. Using Boolean algebra, simplify the following expressions:

(a) $(\overline{A} + B)(A + C)$ **(b)** $A\overline{B} + A\overline{B}C + A\overline{B}CD + A\overline{B}CDE$
(c) $BC + \overline{BCD} + B$ **(d)** $(B + \overline{B})(BC + BC\overline{D})$
(e) $BC + (\overline{B} + \overline{C})D + BC$

21. Using Boolean algebra, simplify the following expressions:

(a) $CE + C(E + F) + \overline{E}(E + G)$ **(b)** $\overline{B}\,\overline{C}D + \overline{(B + C + D)} + \overline{B}\,\overline{C}\,\overline{D}E$
(c) $(C + CD)(C + \overline{C}D)(C + E)$ **(d)** $BCDE + BC(\overline{DE}) + (\overline{BC})DE$
(e) $BCD[BC + \overline{D}(CD + BD)]$

Section 4–6 Standard Forms of Boolean Expressions

23. Convert the following expressions to sum-of-product (SOP) forms:

(a) $(C + D)(A + \overline{D})$ **(b)** $A(A\overline{D} + C)$ **(c)** $(A + C)(CD + AC)$

24. Convert the following expressions to sum-of-product (SOP) forms:

(a) $BC + DE(B\overline{C} + DE)$ **(b)** $BC(\overline{C}\,\overline{D} + CE)$ **(c)** $B + C[BD + (C + \overline{D})E]$

25. Define the domain of each SOP expression in Problem 23 and convert the expression to standard SOP form.

26. Convert each SOP expression in Problem 24 to standard SOP form.

36. For each truth table in Table 4–15, derive a standard SOP and a standard POS expression

TABLE 4–15

(a)

A B C	*X*
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	0
1 1 1	1

(b)

A B C	*X*
0 0 0	0
0 0 1	0
0 1 0	0
0 1 1	0
1 0 0	0
1 0 1	1
1 1 0	1
1 1 1	1

(c)

A B C D	*X*
0 0 0 0	1
0 0 0 1	1
0 0 1 0	0
0 0 1 1	1
0 1 0 0	0
0 1 0 1	1
0 1 1 0	1
0 1 1 1	0
1 0 0 0	0
1 0 0 1	1
1 0 1 0	0
1 0 1 1	0
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	0

(d)

A B C D	*X*
0 0 0 0	0
0 0 0 1	0
0 0 1 0	1
0 0 1 1	0
0 1 0 0	1
0 1 0 1	1
0 1 1 0	0
0 1 1 1	1
1 0 0 0	0
1 0 0 1	0
1 0 1 0	0
1 0 1 1	1
1 1 0 0	1
1 1 0 1	0
1 1 1 0	0
1 1 1 1	1

Section 4–9 Karnaugh Map SOP Minimization

45. Reduce the function specified in truth Table 4–16 to its minimum SOP form by using a Karnaugh map.

46. Use the Karnaugh map method to implement the minimum SOP expression for the logic function specified in truth Table 4–17.

47. Solve Problem 46 for a situation in which the last six binary combinations are not allowed.

TABLE 4–16

Inputs			Output
A	*B*	*C*	*X*
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

TABLE 4–17

Inputs				Output
A	*B*	*C*	*D*	*X*
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

13. Implement a logic circuit for the truth table in Table 5–8.

TABLE 5–8

Inputs			Output
A	*B*	*C*	*X*
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

14. Implement a logic circuit for the truth table in Table 5–9.

TABLE 5–9

Inputs				Output
A	*B*	*C*	*D*	*X*
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

15. Simplify the circuit in Figure 5–57 as much as possible, and verify that the simplified circuit is equivalent to the original by showing that the truth tables are identical.

16. Repeat Problem 15 for the circuit in Figure 5–58.

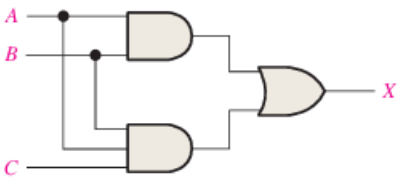


FIGURE 5–57

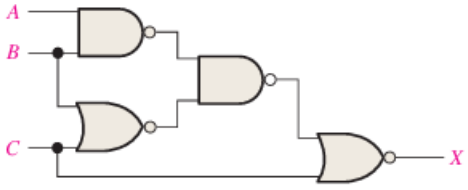


FIGURE 5–58

Section 5–4 Combinational Logic Using NAND and NOR Gates

24. Show how the following expressions can be implemented as stated using only NOR gates:

(a) $X = ABC$ **(b)** $X = \overline{ABC}$ **(c)** $X = A + B$

(d) $X = A + B + \overline{C}$ **(e)** $X = \overline{AB} + \overline{CD}$ **(f)** $X = (A + B)(C + D)$

(g) $X = AB[C(\overline{DE} + \overline{AB}) + \overline{BCE}]$

25. Repeat Problem 24 using only NAND gates.